ECE 332: Engineering Electromagnetics II

Lecture 1

Mark A. Martin

Portland State University

March 31, 2026

Who Am I?

- **Ph.D. Applied Mathematics** from the University of Washington
- **Topics**: numerical analysis, meteorology/climatology, math biology
- Most recently, **primary DSP engineer** at Audio Precision (AP)
- AP designs and manufactures the **most accurate & versatile audio measurement equipment commercially available**
- **Prior work**: materials science, biotechnology, medical physics, IT
- **VP Programming** for the Rose City Astronomers (RCA)
- Most roles involved algorithm and software development
- Please call me Professor, Professor Martin, or Dr. Martin.

RAPS

Professor Josh Mendez (joshmen@pdx.edu) and I (marmart2@pdx.edu) are faculty mentors for **a student club called RA^{2}PS**, which stands for

- "Radio At Portland State"
- or "Radio Astronomy at Portland State."

The club focuses on applications of radio, such as

- communications systems
- radar systems
- HAM radio
- radio astronomy.

Last year, RCA sponsored a Capstone project to construct a 3.2-meter radio telescope, which is now located at the RCA's remote observing site. This year, we're building more scopes through the radio club and starting to explore other projects.

Assignment

Send me a message (via email at marmart2@pdx.edu or Canvas) telling me

- a little bit about yourself,
- what your interests are,
- and how you feel the material in the course will be helpful for you.

A paragraph or two is enough. But feel free to tell me as much as you want. BTW, **I give credit for this assignment**.

(**Due Thursday, April 2 before class**.)

Resources

Required Textbook

Fundamentals of Applied Electromagnetics, 8th Edition by Fawwaz T. Ulaby and Umberto Ravaioli, Pearson Education 2020

Textbook Website

Helpful resources are available at https://em8e.eecs.umich.edu/. These include some problem solutions, interactive illustrations, supplementary articles about applications, and tables of constants and formulas.

The Feynman Lectures on Physics, Volume II

Covers most of the topics we'll discuss and freely available online at https://www.feynmanlectures.caltech.edu/II_toc.html.

Teaching Assistant

Jeff Dinsmore jdinsmor@pdx.edu

Course Material, Coursework, and Grades

Course Material and Grades

- The course will cover **Chapters 6 – 9** in the text.
- Grade will be based on **lab work**, **homework**, and **2 exams**.
  Lab work = 20%, homework = 30%, each exam = 25%.
  Curve applied if needed.
- The **grading rubric** is in the syllabus.
- Important dates will be on the **class calendar in Canvas**.

Homework

- There will be **3 or 4 homework assignments**. Each assignment will be worth roughly 7.5 – 10% of your final grade.
- **Homework assignments** will be submitted electronically via Canvas and will be **due at the beginning of class on the due date**.

Course Material, Coursework, and Grades (Continued)

- I strongly suggest **starting homework assignments as soon as possible**. They will require significant time and effort and waiting until the last minute is a very bad idea.
- Unless you have a related accommodation through the Disability Resource Center (DRC), **late assignments will not be accepted**. If you aren't done with an assignment at the deadline, please turn in the work you have done up to that point.

Exams

- There will be **2 tests of the same length** focusing on the learning outcomes. The final will **not be cumulative**.
- **Final exam**: Tuesday, June 9, 1:30 – 3:20 pm in FAB 48
- **Exams will be in person, on paper**. A calculator approved for the FE exam will be allowed but no other devices. One double-sided page of notes will be allowed as a reference but no other materials.

Course Material, Coursework, and Grades (Continued)

Make-Up Policy

- **Illnesses and other unfortunate events** will be handled on a **case-by-case basis**. Please let me know **as soon as possible** if you experience one of these events and need some flexibility.

Structure of Class

- **Classes** will be held **in person** in FAB 48 and **streamed via Zoom**.
- **Exams** will be held **in person** in FAB 48.
- **Students who attend class in person tend to do better.**
- I will **occasionally pause to answer your questions**. I will address most questions during the pauses.
- We'll have a **5 to 10 minute break** $\approx$ halfway through each class.
- I will be **recording my lectures** via Zoom and **posting the recordings in Canvas and MediaSpace (media.pdx.edu)**.
- **Recordings cannot be shared with anyone outside of our class**.

Structure of Class (Continued)

- There are **3 ways to access lectures**:
  1. attend class in person (**most beneficial**),
  2. connect to class via Zoom during lecture,
  3. watch the recording of a lecture at some later time (**least beneficial**).
- If you connect to class **via Zoom during lecture**:
  - It's best to **ask questions via the chat facility** while I'm lecturing.
  - Please keep your **microphones muted during lecture** but feel free to unmute and respond when I pause for questions.

Submitting Work Electronically

Your **homework will be submitted electronically**. Ways to do this:

- Write your work by hand on paper and then scan or photograph it.

  This is one of the fastest ways. Scans tend to produce better results.

- Use an iPad, reMarkable, Boox, Kindle Scribe, or other device that allows you to create a computer file through writing.

  This is also a fast way.

- Use LaTeX or some other mathematical typesetting software.

  This approach might take the longest.

- Use a Jupyter notebook/JupyterLab, MATLAB Live Script, or another type of document that allows you to combine text, mathematics, code, and graphical output.

In all cases, **please submit your work in PDF, JPEG, or PNG formats and do not submit zip files.**

Submitting Work Electronically (Continued)

Make sure that whatever you submit is **legible** and **well-organized** before submitting it. To do this,

1. **Open the file** and look at it yourself.
   - If it is hard to read, you might have to change settings or lighting conditions and then create a new version of the file.
   - If it is clear, submit to Canvas.
2. **Look at your submission in Canvas** after you have submitted it and **make sure it is still legible**. If it is hard to read in Canvas, you might want to consider recreating the file with different settings or lighting.

If I cannot read your work or it is very disorganized, I might not be able to understand it and you might not receive full credit.

Getting Help

- **Office hours**: Tuesdays and Thursdays from 12:30 – 1:20 pm in FAB 20-10 or via Zoom by appointment. If you need to arrange a Zoom appointment, please contact me in class, via Canvas, or by email (marmart2@pdx.edu) to set up a time.
- **You can always send me questions via email at any time**. You might not receive an answer immediately. But I will try to respond within a reasonable time.
- **I don't usually monitor the "Discussions" section** on the class Canvas site. But **it's a place where you could interact with each other**. Students have set up Discord servers for similar purposes.
- The **Computer Action Team (CAT)** provides help with access to university computing resources, such as MATLAB, or with general questions about computers and software, such as how to install the software you need or how to troubleshoot problems.

More Help

- If you need **disability services**, contact the DRC (drc@pdx.edu), as described in the syllabus. The DRC will then contact me with a list of accommodations to provide during the course.
- If there is some other challenge in your life that is interfering with your ability to focus on your classes, the university might be able to help. One of the points of contact below might be a good place to start.
  - **Basic Needs Hub:** basicneedshub@pdx.edu
  - **Portland State Food Pantry:** psufp.com, pantry@pdx.edu
  - **CARE Program:** askdos@pdx.edu, 503-725-4422
  - **Student Health & Counseling:** askshac@pdx.edu, 503-725-2800

  See the syllabus for more information.
- **Campus Safety**: If you don't feel safe walking around campus, the Campus Public Safety office will provide a campus safety escort anytime, any day (i.e. 24/7). Please call 503-725-4407 to arrange an escort or click here for more information.

Teaching Assistant

- Our teaching assistant (TA), **Jeff Dinsmore** jdinsmor@pdx.edu, will **conduct the labs** and will be available to **help you with any questions**.
- Jeff's **office hours** are TBD.
- You can always **send questions** to Jeff via **Canvas or email** (jdinsmor@pdx.edu) **at any time**. You might not receive an answer immediately. But he will try to respond within a reasonable time.

Note about Large Language Models (LLM), e.g. ChatGPT

How do Large Language Models (LLMs) Work?

Large language models string together words based on statistical predictions of the next most likely word in the current context.

Features of LLMs

- They **do not reason**.
- The output is **generally well-written** and **sounds very convincing**.
- There is **no way to guarantee that the output is true**.
- There is **no way to determine how or why an LLM (or deep-learning model) produced a specific result**.
- The output of an LLM is usually **most accurate for subjects associated with the data used to train it** and **less accurate for other subjects**. (The training data is not specified.)
- Like social media, **LLMs are designed to keep you engaged**. Often, an LLM will flatter you to encourage you to use it more.

Note about LLMs (Continued)

As a result, output from an LLM based on one of the homework problems from our class is likely to be **nonsensical and completely wrong but sound convincing**.

A clear set of examples of poor results from LLMs is in the article

B and not B? That is the answer.

from the September 2023 issue of Math Horizons magazine. A copy of this article is in the Lecture 1 module in Canvas. (YouTube has many examples showing LLMs performing poorly when asked technical questions.)

If LLMs are so bad, why are they used in science and engineering?

- They can produce **novel, surprising results**.
- When an LLM (or deep-learning model) is used in a scientific or engineering context, the results are **thoroughly verified using as many independent approaches as possible**.

Note about LLMs (Continued)

Effects of Using LLMs on Your Brain

MIT Media lab studied the effects of LLM use on people's cognitive skills. The study focused on essay writing, which is the task where LLMs are probably the most useful, and compared 3 groups:

- students with no online resources,
- students using only search engines and Internet resources,
- and students using LLMs.

The **students using LLMs** had

- weakened neural connectivity,
- impaired memory,
- less ownership of their writing,
- and underperformed by neural, linguistic, and behavioral measures.

The paper is available here, an executive summary of the paper is available here, and a YouTube video discussing the paper is available here.

Note about LLMs (Continued)

LLMs and the News

An extensive study of AI assistants conducted by the BBC and the European Broadcasting Network showed that "AI assistants misrepresent news content 45% of the time – regardless of language or territory."

LLMs and Coding

AI-based coding assistants, e.g. copilot and others, **can be helpful**. But you need to **understand code well enough to evaluate (and debug) the results**. Companies that use these tools to accelerate their software development have seen

- a substantial **decrease in development time**
- but also a roughly **40% increase in the bug rate**.

(For example, see this paper about productivity increases and this report regarding the increase in bug rate.)

Note about LLMs (Continued)

My personal opinion is

- It's too soon to rely on these tools.
- Reliably using an LLM requires thorough evaluation, which requires a solid foundation of knowledge, like the one we're working to develop in this course.

<u>**Bottom line**</u>: I strongly suggest you **avoid trying to use an LLM to solve problems in our class**.

Review of Ideas from ECE 331

ECE 331 developed the **definitions** and **mathematical infrastructure** needed to understand electromagnetic phenomena. You learned that 4 vector fields –

- the **electric field intensity** $\mathbf{E}$ (V/m),
- the **electric flux density** $\mathbf{D} = \epsilon\mathbf{E}$ (C/m^2),
- the **magnetic field intensity** $\mathbf{H}$ (A/m),
- and the **magnetic flux density** $\mathbf{B} = \mu\mathbf{H}$ (T)

determine the behavior of this type of phenomena and that **vector calculus is the mathematical language appropriate for describing them**. Here, ϵ (F/m) is the **electrical permittivity** and μ (H/m) is the **magnetic permeability** of the medium.

The fact that **light is electromagnetic** was also discussed (more on this later) and you explored several applications, most notably the nature of **transmission lines** and factors important in their use and design.

Review of Ideas from ECE 331 (Continued)

The fields determine the **force on a test charge** q. Specifically, the force due to the electric fields is

$$\mathbf{F} = q\mathbf{E} \tag{1}$$

and the force on a moving charge with velocity $\mathbf{u}$ due to the magnetic fields is

$$\mathbf{F} = q\mathbf{u} \times \mathbf{B}. \tag{2}$$

Electric and magnetic fields due to different sources can be combined via **superposition** and can be considered independently. So, the overall force on a test charge q with velocity $\mathbf{u}$ is the **Lorentz Force**

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \tag{3}$$

(Note that superposition in this context means exactly the same thing as superposition in signal processing.)

Review of Ideas from ECE 331 (Continued)

Maxwell's Equations

Experimentation and theoretical work in the 19th century established that the fields **E**, **D**, **H**, and **B** satisfy and are determined by **Maxwell's equations** and appropriate **boundary conditions**.

When **all electrical charges** that are not part of a current **are fixed in place** and **all currents are steady**, the equations describing the electric fields **E** and **D** are independent of the equations describing the magnetic fields **H** and **B**.

Review of Ideas from ECE 331 (Continued)

In this case, Maxwell's equations take the form

$$\nabla \cdot \mathbf{D} = \rho_v \longleftrightarrow \oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \text{(Gauss' Law)} \tag{4}$$

$$\nabla \times \mathbf{E} = \mathbf{0} \longleftrightarrow \oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \quad \text{(Kirchhoff's Law)} \tag{5}$$

(electrostatics)

where ρ_v is the **charge density** ($\mathrm{C/m^3}$) and Q is the **charge enclosed in the surface** S, and

$$\nabla \cdot \mathbf{B} = 0 \longleftrightarrow \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \text{(Gauss' Law for Magnetism)} \tag{6}$$

$$\nabla \times \mathbf{H} = \mathbf{J} \longleftrightarrow \oint_C \mathbf{H} \cdot d\mathbf{l} = I \quad \text{(Ampère's Law)} \tag{7}$$

(magnetostatics)

where $\mathbf{J} = \rho_v \mathbf{u}$ is the **current density** ($\mathrm{A/m^2}$) and I is the **current along the contour** C. These determine **E**, **D**, **H**, and **B** in the static case.

Review of Ideas from ECE 331 (Continued)

Coulomb's Law

A 2nd way to determine $\mathbf{E}$ (and $\mathbf{D} = \epsilon\mathbf{E}$) is to calculate it directly from the charge distribution. According to Coulomb's law,

$$\mathbf{E} = \frac{1}{4\pi\epsilon}\sum_{i=1}^{N}\frac{q_i(\mathbf{R}-\mathbf{R}_i)}{|\mathbf{R}-\mathbf{R}_i|^3} \tag{8}$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon}\int_{l'}\hat{\mathbf{R}}'\frac{\rho_l dl'}{R'^2}, \quad \mathbf{E} = \frac{1}{4\pi\epsilon}\int_{S'}\hat{\mathbf{R}}'\frac{\rho_s ds'}{R'^2}, \quad \mathbf{E} = \frac{1}{4\pi\epsilon}\int_{\mathcal{V}'}\hat{\mathbf{R}}'\frac{\rho_v d\mathcal{V}'}{R'^2}, \tag{9}$$

for a collection of point charges and line, surface, and volume charge distributions, respectively. In the first equation, $\mathbf{R}$ is the location of the point where the field is being evaluated. In the integrals, $\mathbf{R}' = R'\hat{\mathbf{R}}'$ is the vector from the line, surface, or volume differential where charge is located to the point where the field is being evaluated, $R' = |\mathbf{R}'|$, and $\hat{\mathbf{R}}'$ is the unit vector pointing in the direction of $\mathbf{R}'$.

Review of Ideas from ECE 331 (Continued)

Biot-Savart Law

A 2nd way to determine $\mathbf{H}$ (and $\mathbf{B} = \mu\mathbf{H}$) is to calculate it directly from the distribution of currents. According to the Biot-Savart law,

$$\mathbf{H} = \frac{1}{4\pi}\int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}, \quad \mathbf{H} = \frac{1}{4\pi}\int_S \frac{\mathbf{J}_s \times \hat{\mathbf{R}}}{R^2}\, ds, \quad \mathbf{H} = \frac{1}{4\pi}\int_{\mathcal{V}} \frac{\mathbf{J} \times \hat{\mathbf{R}}}{R^2}\, d\mathcal{V}, \quad (10)$$

for line, surface, and volume currents, respectively. Here, $\mathbf{R} = R\hat{\mathbf{R}}$ is the vector from the line, surface, or volume differential where current is flowing to the point where the field is being evaluated, $R = |\mathbf{R}|$, and $\hat{\mathbf{R}}$ is the unit vector in the direction of $\mathbf{R}$.

Review of Ideas from ECE 331 (Continued)

Potentials

In the electrostatic case, the electric field intensity can be expressed as

$$\mathbf{E} = -\nabla V, \tag{11}$$

where V is the **electric potential**. For 2 points in a circuit, this is the voltage between the points and is the amount of potential energy needed to move a unit charge from one point to the other. Mathematically, it's consistent with the fact that $\nabla \times (\nabla V) = \mathbf{0}$ for any scalar function V.

The potential can be calculated from the charge distribution according to one of the formulas

$$V = \frac{1}{4\pi\epsilon} \sum_{i=1}^{N} \frac{q_i}{|\mathbf{R} - \mathbf{R}_i|}, \tag{12}$$

$$V = \frac{1}{4\pi\epsilon} \int_{l'} \frac{\rho_l}{R'}\, dl', \quad V = \frac{1}{4\pi\epsilon} \int_{S'} \frac{\rho_s}{R'}\, ds', \quad V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'}\, d\mathcal{V}'. \tag{13}$$

This provides a 3rd way to determine $\mathbf{E}$ (and $\mathbf{D} = \epsilon\mathbf{E}$).

Review of Ideas from ECE 331 (Continued)

The **vector magnetic potential A** (Wb/m) is defined by

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{14}$$

This relationship guarantees that the magnetic flux density **B** satisfies Gauss' Law for Magnetism since

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \tag{15}$$

for any vector field **A**. A viable assumption regarding **A** is that

$$\nabla \cdot \mathbf{A} = 0. \tag{16}$$

Under this assumption,

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}, \quad \text{(Poisson's equation)} \tag{17}$$

which implies

$$\mathbf{A} = \frac{\mu}{4\pi} \int_{l'} \frac{\mathbf{J}}{R'}\, dl', \quad \mathbf{A} = \frac{\mu}{4\pi} \int_{S'} \frac{\mathbf{J}}{R'}\, ds', \quad \mathbf{A} = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}}{R'}\, d\mathcal{V}' \tag{18}$$

and provides a 3rd way to calculate **B** (and $\mathbf{H} = \frac{1}{\mu}\mathbf{B}$).

Review of Ideas from ECE 331 (Continued)

More Electrostatics

- Materials are classified as **conductors** or **dielectrics** (insulators).
- The electric potential V also satisfies Poisson's equation

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}. \tag{19}$$

- The **electrostatic energy density** for a dielectric medium is

$$w_e = \frac{1}{2}\epsilon E^2 \quad (\mathrm{J/m^3}), \tag{20}$$

  where $E = |\mathbf{E}|$. It is the electrostatic potential energy per unit volume.
- **Ohm's law** for a conducting medium can be written in the form

$$\mathbf{J} = \sigma \mathbf{E} \quad (\mathrm{A/m^2}), \tag{21}$$

  where σ (S/m) is the **conductivity** of the medium.

Review of Ideas from ECE 331 (Continued)

- **Boundary conditions** apply to interfaces between media with different properties. They specify how the components of **E**, **D**, and **J** that are normal and tangential to an interface on the two sides of the interface are related.

More Magnetostatics

- The **magnetic force for a closed circuit of contour** C carrying current I is

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \tag{22}$$

- The **magnetic torque** (i.e. twisting force) on a loop is

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\mathrm{N} \cdot \mathrm{m}), \tag{23}$$

where the **magnetic moment** of the loop is

$$\mathbf{m} = \hat{\mathbf{n}} N I A \quad (\mathrm{A} \cdot \mathrm{m}) \tag{24}$$

with $\hat{\mathbf{n}}$ = unit surface normal of the loop, N = number of turns, I = current through the wire, A = cross-sectional area of the wire.

Review of Ideas from ECE 331 (Continued)

- Materials are classified as **diamagnetic**, **paramagnetic**, or **ferromagnetic** based on their crystalline structure and their response to an external magnetic field. Atoms for paramagnetic and ferromagnetic materials have permanent magnetic moments whereas atoms in diamagnetic materials do not. **Ferromagnetic materials exhibit hysteresis** in the relationship between B and H.
- The **magnetic energy density** in any medium with a magnetic field **H** is

$$w_m = \frac{1}{2}\mu H^2 \tag{25}$$

- **Boundary conditions** apply to interfaces between media with different properties. They specify how the components of **B** and **H** that are normal and tangential to an interface on the two sides of the interface are related.

Quite a few useful devices can be analyzed and understood from the perspectives of electro- and magnetostatics.

Review of Ideas from ECE 331 (Continued)

Phasors

Phasors are **complex numbers or complex-valued functions** that provide **a compact way to represent and compare real-valued sinusoidal quantities that have the same angular frequency** ω.

Some Uses of Phasors

1. Phasors are used to represent the **steady-state sinusoidal voltages and currents in ac circuits** and to determine the **various types of power associated with these circuits**.

   In this case, a **voltage or current** in a circuit can be written as

   $$z(t) = A\cos(\omega t + \phi_0) = \mathfrak{Re}\left[Ae^{j(\omega t+\phi_0)}\right] \tag{26}$$

   and represented as the **phasor**

   $$\tilde{Z} = Ae^{j\phi_0}. \tag{27}$$

   In this case, the phasor is a **complex number**.

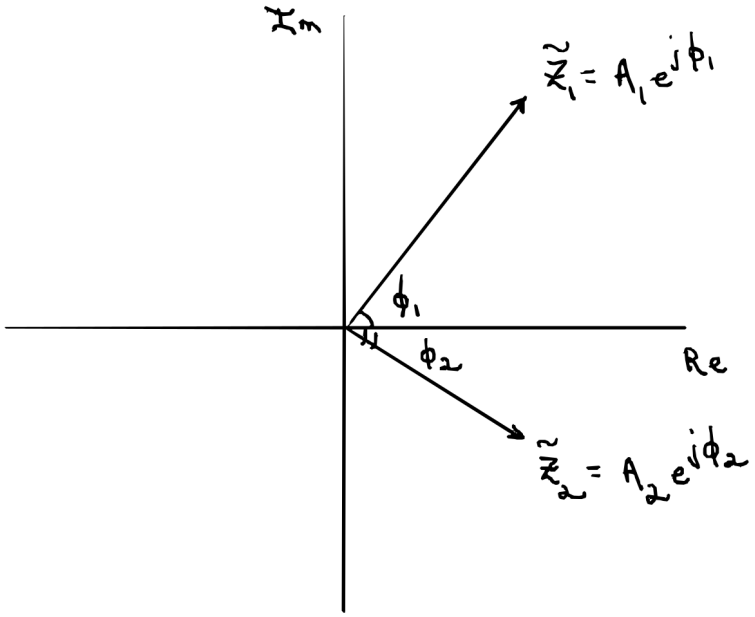
Review of Ideas from ECE 331 (Continued)

If the phasor $\tilde{Z}$ is known, then the **time-dependent quantity** can be determined via the formula

$$z(t) = \mathfrak{Re}\left[\tilde{Z}e^{j\omega t}\right] = \mathfrak{Re}\left[Ae^{j\phi_0}e^{j\omega t}\right]. \tag{28}$$

Typically, **two or more sinusoidal voltages or currents** in an ac circuit are of interest. These will have the **same angular frequency** but will **differ in phase and amplitude**.

A **phasor diagram** shows the phasors of the voltages and/or currents and is a succinct way of representing the steady-state ac behavior of a circuit.



Review of Ideas from ECE 331 (Continued)

Example

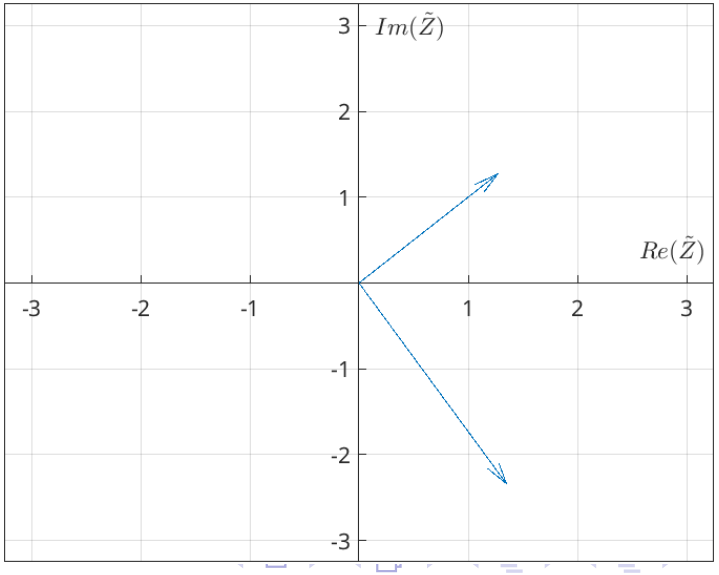
Consider the real-valued sinusoidal quantities

$$z_1(t) = 2\cos(32\pi t + \pi/4) \text{ and } z_2(t) = 3\cos(32\pi t - \pi/3). \quad (29)$$

The corresponding phasors are

$$\tilde{Z}_1 = 2e^{j\pi/4} = 2(\cos(\pi/4) + j\sin(\pi/4)) = \sqrt{2} + j\sqrt{2}$$
$$\tilde{Z}_2 = 3e^{-j\pi/3} = 3(\cos(-\pi/3) + \sin(-\pi/3)) = \frac{3}{2}(1 - j\sqrt{3}). \quad (30)$$

In this case, the phasor diagram is the one shown.



Review of Ideas from ECE 331 (Continued)

2. Phasors can also be used to represent **sinusoidal traveling waves** that vary in space as well as time.

   For a **sinusoidal plane wave** (i.e. a wave with planar wavefronts) **traveling in the x-direction**, the quantity that oscillates can be written in the form

$$z(t,x) = A\cos(\omega t + \beta x + \phi_0) = \mathfrak{Re}\left[Ae^{j(\omega t+\beta x+\phi_0)}\right] \tag{31}$$

   and represented as the phasor

$$\tilde{Z}(x) = Ae^{j(\beta x+\phi_0)}. \tag{32}$$

   In this case, the phasor is a **complex-valued function of the spatial variable** x and the real part of the phasor $\mathfrak{Re}[\tilde{Z}(x)] = z(0,x)$ is the waveform at time $t = 0$. As before, if the phasor $\tilde{Z}(x)$ is known, the time-dependent quantity can be determined as

$$z(t,x) = \mathfrak{Re}\left[\tilde{Z(x)}e^{j\omega t}\right] = \mathfrak{Re}\left[Ae^{j(\beta x+\phi_0)}e^{j\omega t}\right]. \tag{33}$$

Review of Ideas from ECE 331 (Continued)

Example

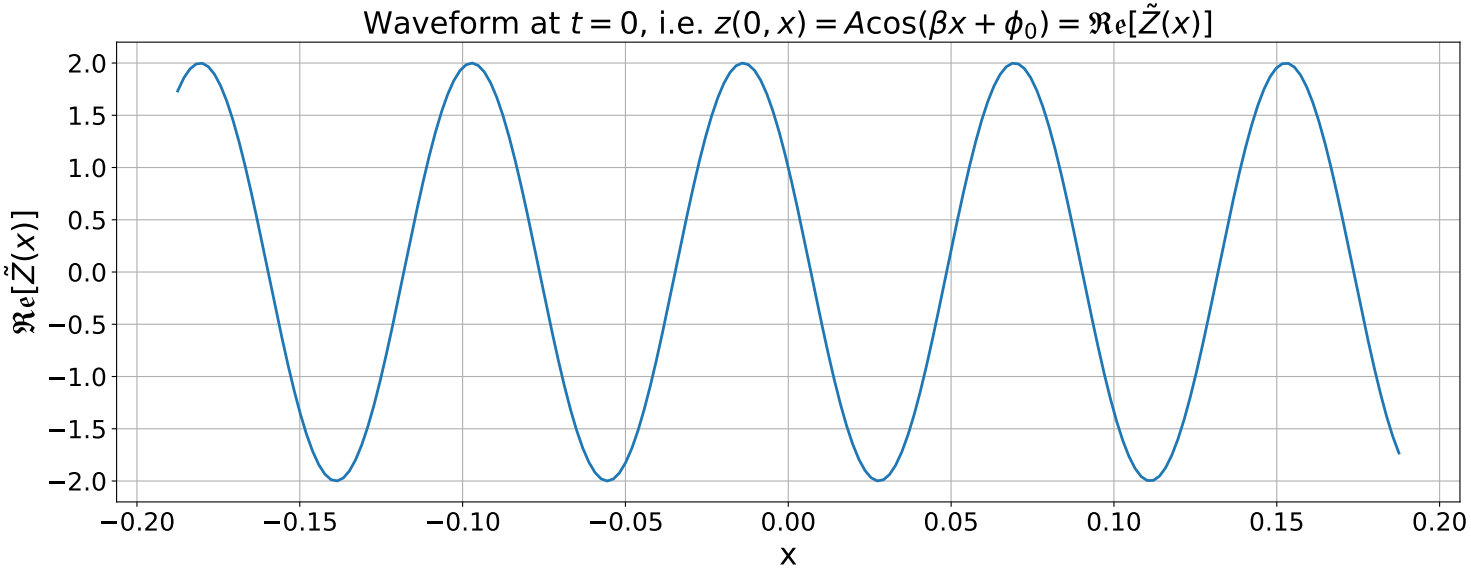
The sinusoidal plane wave

$$z(t,x) = 2\cos\left(10\pi t + 24\pi x + \pi/3\right), \tag{34}$$

which is traveling in the x-direction, has the phasor

$$\tilde{Z}(x) = 2e^{j(24\pi x + \pi/3)}. \tag{35}$$

The real part of the phasor is the waveform at $t = 0$. Here, $A = 2$, $\omega = 10\pi$, $\beta = 24\pi$, $\phi_0 = \frac{\pi}{3}$, and the real part of the phasor is below.



Review of Ideas from ECE 331 (Continued)

3. Lastly, phasors can represent **sinusoidal traveling waves that are exponentially attenuated in the direction of travel**.

   Under these circumstances, for an exponentially attenuated sinusoidal plane wave traveling in the x-direction, the quantity that oscillates can be written in the form

$$z(t,x) = Ae^{-\alpha x}\cos(\omega t + \beta x + \phi_0) = \mathfrak{Re}\left[Ae^{-\alpha x}e^{j(\omega t+\beta x+\phi_0)}\right], \quad (36)$$

   which has the phasor

$$\tilde{Z}(x) = Ae^{-\alpha x}e^{j(\beta x+\phi_0)}. \quad (37)$$

   In this case, the phasor is again a **complex-valued function of the spatial variable** x and the real part of the phasor $\mathfrak{Re}[\tilde{Z}(x)]$ is the waveform at time $t = 0$.

   As in the previous cases,

$$z(t,x) = \mathfrak{Re}\left[\tilde{Z}(x)e^{j\omega t}\right] = \mathfrak{Re}\left[Ae^{-\alpha x}e^{j(\beta x+\phi_0)}e^{j\omega t}\right]. \quad (38)$$

Review of Ideas from ECE 331 (Continued)

Example

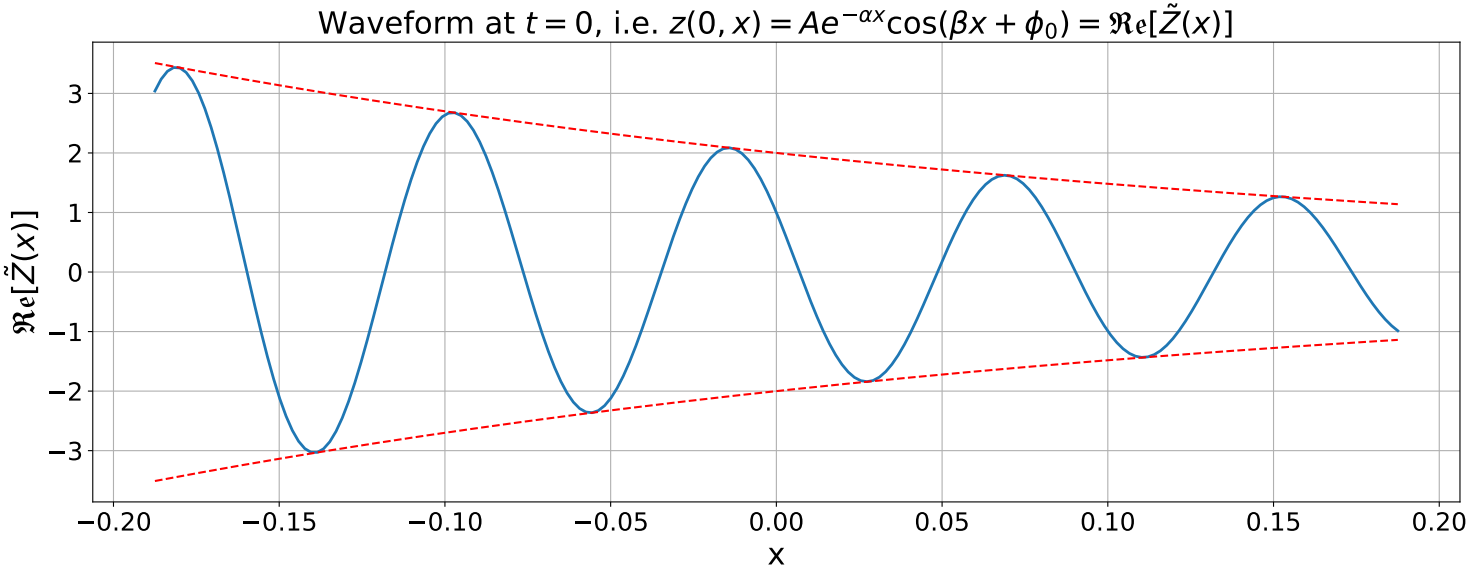
The exponentially attenuated sinusoidal plane wave

$$z(t,x) = 2e^{-3x}\cos\left(10\pi t + 24\pi x + \pi/3\right), \tag{39}$$

which is traveling in the x-direction, has the phasor

$$\tilde{Z}(x) = 2e^{-3x}e^{j(24\pi x+\pi/3)}. \tag{40}$$

The real part of the phasor is the waveform at $t = 0$. Here, $A = 2$, $\omega = 10\pi$, $\alpha = 3$, $\beta = 24\pi$, $\phi_0 = \frac{\pi}{3}$, and $\mathfrak{Re}[\tilde{Z}(x)]$ is shown below.



Maxwell's Equations for Time-Varying Fields

Moving charges create magnetic fields and varying currents exert a force on charges. So, when charges are allowed to move and currents are allowed to vary, the electric and magnetic fields are coupled and Maxwell's equations become

$$\nabla \cdot \mathbf{D} = \rho_v \longleftrightarrow \oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \text{(Gauss' Law)} \tag{41}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \longleftrightarrow \oint_C \mathbf{E} \cdot d\mathbf{l} = -\oint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad \text{(Faraday's Law)} \tag{42}$$

$$\nabla \cdot \mathbf{B} = 0 \longleftrightarrow \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \text{(Gauss' Law for Magnetism)} \tag{43}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \longleftrightarrow \oint_C \mathbf{H} \cdot d\mathbf{l} = \oint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s} \quad \text{(Ampère's Law)} \tag{44}$$

In theory, to determine the fields **E**, **D**, **B**, and **H** using these equations would mean solving them all simultaneously. At this point, we'll focus on the investigations that led to these equations.

Homework

Read Chapter 6.